



Jenkinsfile

Introduction

A **Jenkinsfile** is a script that defines a **Jenkins Pipeline**, enabling **Continuous Integration (CI)** and **Continuous Deployment (CD)**.

- What is a Jenkinsfile?
- Why is it important in modern software development?
- How can you create and use a Jenkinsfile in Jenkins?

This guide provides an **in-depth understanding of Jenkinsfile and its usage**.

Section 1: Learn

What is a Jenkinsfile?

A **Jenkinsfile** is a **text file** that contains a **pipeline script**. It is used to **define, manage, and execute Jenkins Pipelines as code**.

Why Use a Jenkinsfile?

Feature	Benefit
Pipeline as Code	Stores pipeline configuration in version control (Git)
Automation	Automates CI/CD workflows
Reusability	Standardized pipeline scripts for multiple projects
Parallel Execution	Supports running multiple jobs simultaneously



Feature	Benefit
Error Handling	Provides detailed logs and notifications

Types of Jenkinsfile Pipelines

1. **Declarative Pipeline** – Uses a structured and easy-to-read syntax.
2. **Scripted Pipeline** – Uses Groovy scripting for more flexibility.

Interesting Fact

Jenkinsfile was **introduced in Jenkins 2.0** to **bring CI/CD pipelines under version control**, similar to how developers manage source code in Git.

Section 2: Practice

1. Creating a Jenkinsfile (Declarative Pipeline)

Step 1: Create a New Jenkins Pipeline Job

1. Open **Jenkins Dashboard** → Click **New Item**.
2. Enter a **Pipeline Name** → Select **Pipeline** → Click **OK**.
3. Scroll to **Pipeline Definition** → Select **Pipeline script from SCM**.
4. Enter the **Git Repository URL** where the Jenkinsfile is stored.

Step 2: Define a Jenkinsfile

Create a file named **Jenkinsfile** inside your **Git repository**.

```
pipeline {  
  agent any  
  stages {  
    stage('Checkout') {
```



```
    steps {  
        git 'https://github.com/your-repo.git'  
    }  
}  
  
stage('Build') {  
    steps {  
        sh 'mvn clean install'  
    }  
}  
  
stage('Test') {  
    steps {  
        sh 'mvn test'  
    }  
}  
  
stage('Deploy') {  
    steps {  
        echo 'Deploying application...'  
    }  
}  
}
```

Step 3: Save and Run the Pipeline

1. Click **Save** → **Build Now**.
2. Check the **Console Output** for execution logs.

Why is this useful?



- Allows developers to **automate CI/CD workflows in a structured way**.
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2. Creating a Jenkinsfile (Scripted Pipeline)

A **Scripted Pipeline** offers more flexibility using **Groovy scripting**.

Example Scripted Jenkinsfile

```
node {  
    stage('Checkout') {  
        git 'https://github.com/your-repo.git'  
    }  
    stage('Build') {  
        sh 'mvn clean install'  
    }  
    stage('Test') {  
        sh 'mvn test'  
    }  
    stage('Deploy') {  
        echo 'Deploying application...'  
    }  
}
```

Why is this important?

- Provides **full control over Jenkins Pipeline execution**.
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3. Using Parameters in a Jenkinsfile

Jenkinsfiles support **dynamic parameters** for user inputs.



Example: Using Parameters

```
pipeline {
  agent any
  parameters {
    string(name: 'ENV', defaultValue: 'dev', description: 'Select Environment')
  }
  stages {
    stage('Build') {
      steps {
        echo "Building for ${params.ENV} environment"
      }
    }
  }
}
```

Why is this useful?

- Allows **customizable builds** based on user inputs.
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Section 3: Know More

Frequently Asked Questions

1. **Where should I store a Jenkinsfile?**
 - Store it in the **root directory** of your **Git repository**.
2. **Can I use multiple Jenkinsfiles in a single project?**
 - Yes, you can define **different Jenkinsfiles** for **different branches**.
3. **How do I debug errors in a Jenkinsfile?**



- Check the **Console Output** in Jenkins for detailed error logs.
 - 4. **What is the difference between Declarative and Scripted Pipelines?**
 - **Declarative Pipelines** are easier to read, while **Scripted Pipelines** offer more flexibility.
 - 5. **Can Jenkinsfiles be used with cloud platforms?**
 - Yes! Jenkinsfiles can be used to deploy applications to **AWS, Azure, Kubernetes, and Docker**.
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Conclusion

Jenkinsfiles **enable automated, version-controlled, and reusable CI/CD workflows.**

Start by **creating a basic Jenkinsfile, adding parameters, and integrating with Git**, and soon you'll be **optimizing your software deployment processes efficiently!**